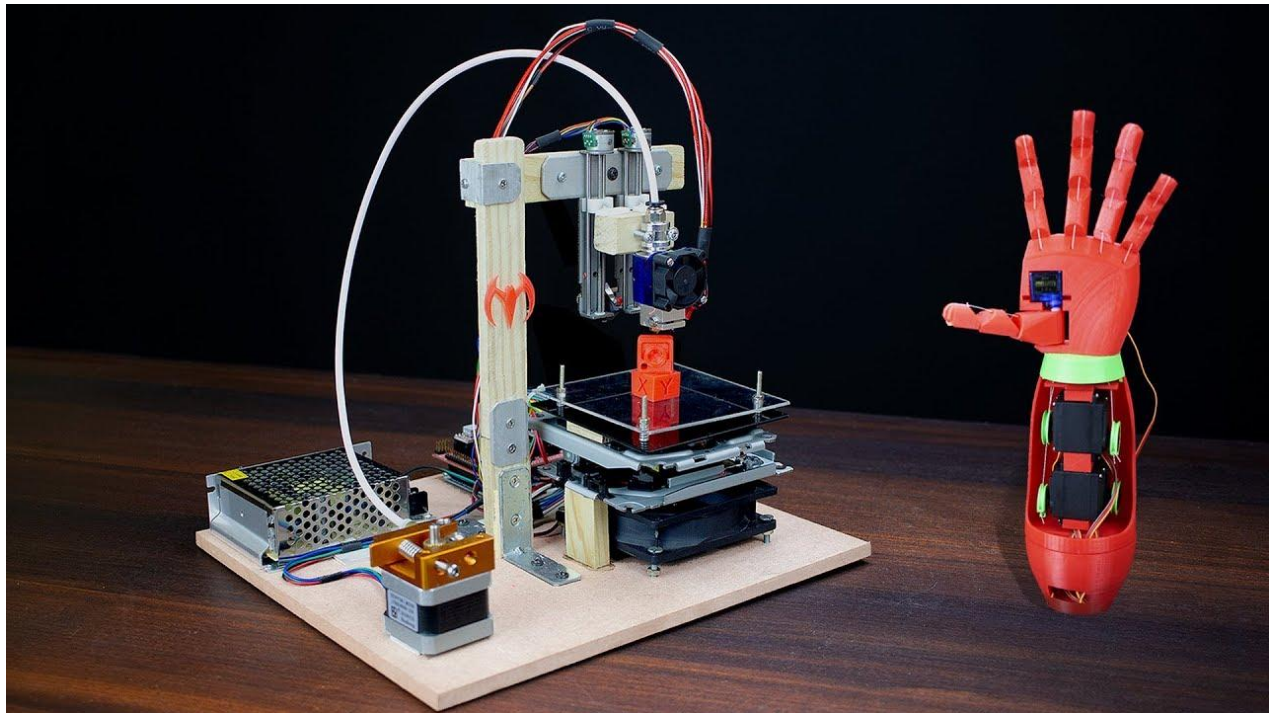


3D PRINTING + ARDUINO COURSE



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
16 CLASSES (1 HOUR PER CLASS)**

**MODE:
OFFLINE**

It is a process of making 3-dimensional solid object from a digital file 3D is shorthand for three-dimensional. When you print a page on a printer, there are only two dimensions: the front of the page and the back of the page. Three-dimensional printing adds a third dimension, volume. 3D printing uses a printer to create three-dimensional objects, for example, a cup or doll or phone case and replaces the wax models. This technology is currently used in many fields such as health, construction, food, engineering, aerospace, etc.

We complete on an average of 8-10 models as per the progress and speed of child in understanding the technology and its techniques.

Level 1: Basics of Arduino & 3D Design (Ages 10–13)

1. LED Blink & Traffic Light Simulation

- Learn about digital output using digitalWrite
- Hands-on: Program blinking LED & traffic light pattern

2. Intro to Tinkercad & Keychain Design

- Learn to design using kid-friendly software
- Hands-on: Design and print a personalized name keychain



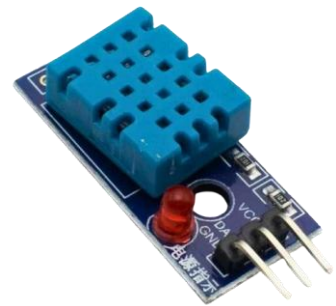
AUTODESK®
TINKERCAD®

3. Button Input & Buzzer Output (If-Else Logic)

- Understand conditional logic in Arduino
- Hands-on: Use button to activate buzzer

4. Phone Stand Modeling

- Explore dimensions and functional modeling
- Hands-on: Design and print a functional phone stand



5. Temperature & Humidity Sensor (DHT11)

- Use sensor to monitor environmental data
- Hands-on: Display temperature & humidity on Serial Monitor

6. Mini House Design

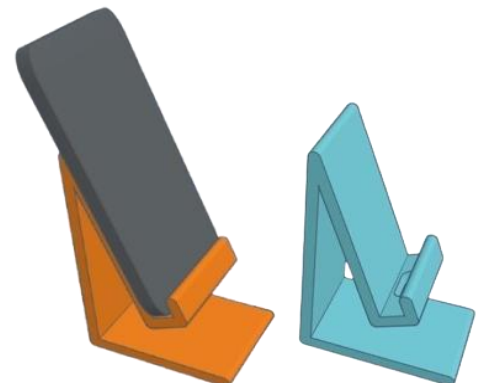
- Understand basic architecture modeling
- Hands-on: Design and print a house with walls, doors, and roof

7. Ultrasonic Sensor for Distance Measurement

- Measure real-time distance using sensors
- Hands-on: Display distance using Arduino

8. Rocket or Fish Shape Modeling

- Learn symmetry and shape merging



- Hands-on: Design and print a rocket or fish model

9. Servo Motor with Potentiometer

- Use analog input to control rotation
- Hands-on: Build a circuit to control servo using potentiometer



10. Flower Pot Design with Text Engraving

- Learn how to hollow objects and add text
- Hands-on: Design and print a flower pot with engraved name

11. BO Motor Control using L293D Driver

- Understand motor driving logic and direction control
- Hands-on: Control BO motor using L293D and Arduino



12. Robot Chassis Design for Sensors

- Learn to create practical enclosures and slots
- Hands-on: Design a robot chassis with mounts and sensor slots

13. Obstacle Avoidance Car Build

- Combine ultrasonic sensor and motor logic
- Hands-on: Program and test a car that avoids obstacles

14. Clock Frame or Otto Shell Design

- Learn curved shapes and functional structure modeling
- Hands-on: Design and print frame for clock or Otto bot

15. Otto Bot or Clock Final Testing

- Integrate servo, motor, and structure
- Hands-on: Final testing and troubleshooting

16. Final 3D Model & Review Presentation

- Complete custom model and prepare explanation
- Hands-on: Print model and present concept and function



Level 2: Smart Robotics & 3D Fusion Design (Ages 13–15)

1. LDR + LED Indicators

- Detect light sensitivity for line-following
- Hands-on: Use LDRs to trigger LEDs

2. Intro to Fusion 360 + Multi-Part Modeling

- Navigate advanced CAD interface
- Hands-on: Design and print multi-part object



3. Ultrasonic + Servo Motor Radar

- Use servo to rotate ultrasonic sensor
- Hands-on: Simulate radar scanning

4. Parametric Design & Constraints

- Use dimensions and constraints in Fusion
- Hands-on: Design and print a snap-fit box



5. IR Remote + Servo/LED Control

- Control outputs using IR remote
- Hands-on: Move servo or blink LED wirelessly

6. Loft & Fillet Tool Practice

- Smooth surfaces and blend shapes
- Hands-on: Create curved casings or enclosures

7. Dual Motor + IR Remote Movement

- Control robot direction using remote
- Hands-on: Build a remote-operated bot

8. Precision Measurement & Fit Design

- Take accurate dimensions
- Hands-on: Design a tight-fit mobile case



9. Wall-E Robot – Chassis & Sensors

- Assemble robot base and place IR/ultrasonic sensors
- Hands-on: Build Wall-E body for tracking application

10. Blender Basics + 3D Keychain

- Explore interface and create curved shapes
- Hands-on: Make a Blender-based 3D name keychain



11. Wall-E Robot – Line Following Logic

- Apply motor and IR sensor logic
- Hands-on: Test and refine robot path-following

12. Interlocking Object Design

- Create multi-part structures that fit together
- Hands-on: Print and assemble a modular pen holder

13. Fire-Fighting Robot – Flame Detection

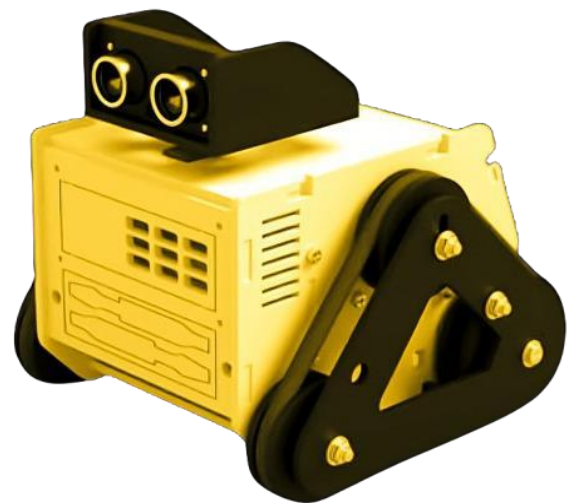
- Use flame sensor to detect fire
- Hands-on: Build system to alert and respond

14. Snap-Fit Enclosure Design

- Create a closed container using joint techniques
- Hands-on: Design and print a snap-fit case

15. Fire-Fighting Robot – Final Test

- Add fan and complete full robot
- Hands-on: Test firefighting logic and mobility



16. Freestyle 3D Design Challenge

- Solve a daily-life problem using 3D design
- Hands-on: Design, print and present a useful household/tool item

Level 3: Automation, AI & Advanced CAD (Ages 15–18)

1. PIR Sensor Motion Detection System

- Detect movement to trigger outputs
- Hands-on: Create a motion-activated security setup



2. Advanced Gear & Mechanical Design

- Understand gear mechanics and tolerances
- Hands-on: Print and test working gear mechanism



3. Relay Module for AC/DC Control

- Safely switch high-voltage devices
- Hands-on: Control fan/light using relay and Arduino

4. Strength-Oriented Design Principles

- Learn about infill, shell, and wall thickness
- Hands-on: Design and print a durable key holder

5. Bluetooth + App Control via HC-05

- Use smartphone to control Arduino outputs
- Hands-on: Build app-connected robot or device

6. Interlocking Modular Storage Systems

- Explore stackable design techniques
- Hands-on: Design and print parametric storage drawers

7. Servo Finger Controlled by Potentiometer

- Simulate joint movement using rotation control
- Hands-on: Build one-finger prototype with real-time control

8. Snap-Fit + Hinge-Based Enclosure Design

- Combine opening/closing mechanics with fit
- Hands-on: Print a working hinged enclosure



9. Bionic Hand – Flex Sensor Input

- Detect finger bends using sensors
- Hands-on: Build one-finger flex response logic

10. Robotic Joint Design for Arms

- Understand mounts, rotation, and servo fitting
- Hands-on: Design and print robotic arm part

11. Bionic Hand – Servo Integration

- Add motion to bionic finger
- Hands-on: Link flex input to servo response

12. Threaded Cap & Container Design

- Model screw threads for locking mechanisms
- Hands-on: Design and print cap + container set

13. Gas/Smoke Detection System with MQ Sensor

- Detect harmful gases, trigger safety alert
- Hands-on: Build and test gas detection logic

14. Final Advanced 3D Design Planning

- Brainstorm and sketch solution-based product
- Hands-on: Start modeling final design

15. Home Automation with Relay & Sensors

- Integrate Bluetooth, PIR, and Relay modules
- Hands-on: Build a multi-functional smart home setup

16. Final Project Showcase & Presentation

- Display complete Arduino + 3D integrated project
- Hands-on: Present features, working, and design story

